

GENEMIND SAMPLES BATCH – 2 (MAY) 16 SAMPLES COMPARISSION

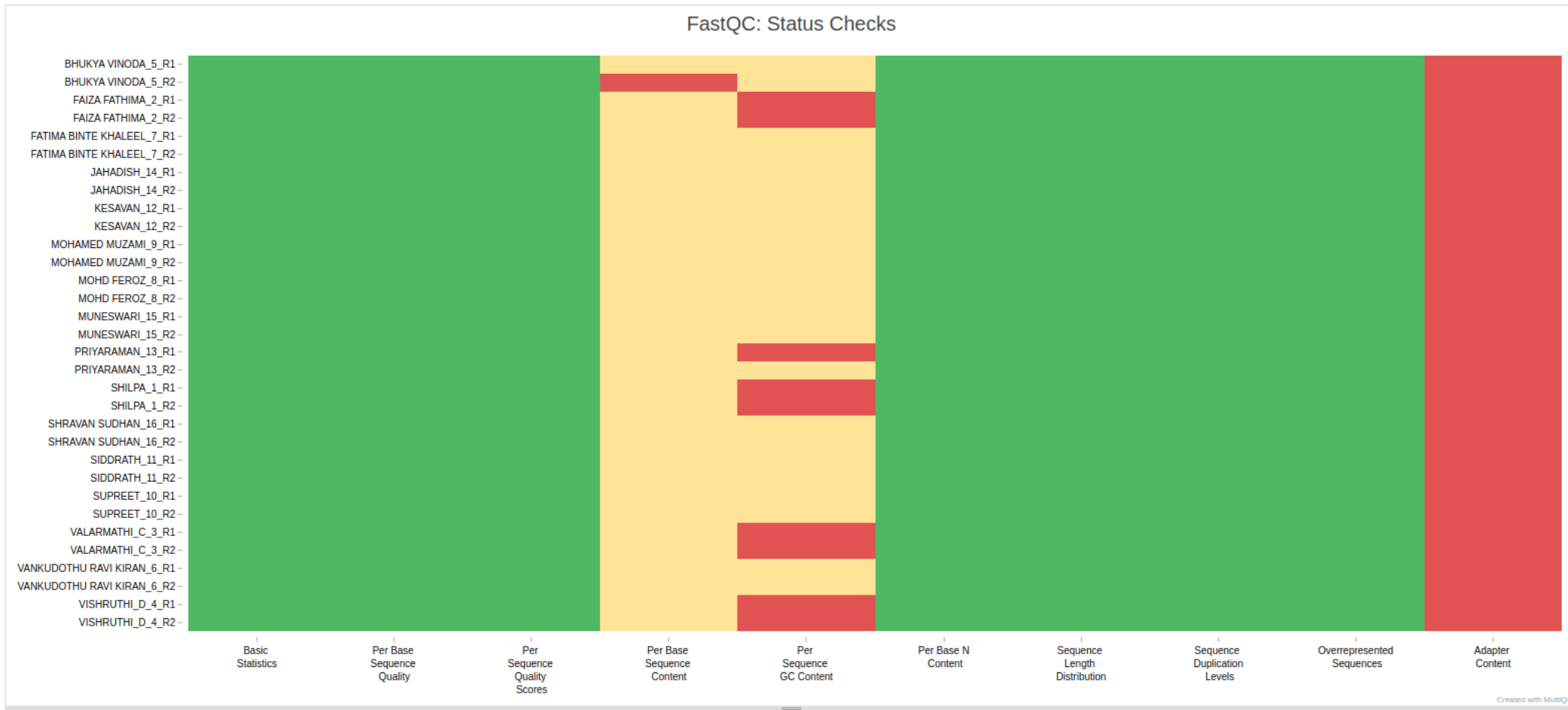
GENEMIND BATCH 16 SAMPLES : (TOTAL SIZE = 124.5 GB)

STEP	INPUT FILE	PROCESS	TOOLS USED	DURATION OF TIME	DATA SIZE
1	FASTQ (R1, R2)	MULTIQC ANALYSIS	MULTIQC, FASTQC	2 HRS	15 MB
2	FASTQ (R1, R2)	FASTP ANALYSIS	FASTP	1 HRS	120 GB
3	TRIMMED FASTQ (R1, R2)	MULTIQC ANALYSIS	MULTIQC, FASTQC	2 HRS	15 MB
4	TRIMMED FASTQ (R1, R2)	HS METRICS ANALYSIS	SENTIEON	4 HRS	100 GB
5	TRIMMED FASTQ (R1, R2)	SECONDARY ANALYSIS	SENTIEON	6 HRS	130 GB
6	BAM FILES	CHROMOSOME COVERAGE	INDEXCOV, PANDA	1 HRS	10 GB
7	VCF, BAM AND BED	SAMPLE SEX AND COVERAGE ANALYSIS	VARSEQ VS CLINICAL COVERAGE AND SAMPLE STASTICS	3 DAYS	8 GB
8	VCF	VARAINT DEPTH ANALYSIS	VARSEQ	1 DAYS	4 GB
9	BAM AND VCF	TERTIARY ANALYSIS AND REPORT PREPARATION	VARSEQ	3 DAYS	10 GB
TOTAL				~ 6 DAYS	~ 520 GB

STEP : 1 – MULTIQC ANALYSIS BEFORE TRIMMING

- INPUT FILES : R1, R2 FASTQ FILES
- TOOL : FASTQC, MULTIQC
- TIME TAKEN: (2 HRS FOR 16 SAMPLES)
- DATA SIZE : 15 MB (FINAL OUTPUT)

FASTQC RESULT BEFORE TRIMMING:



STEP 2 : FASTP (ADAPTER TRIMMING OF FASTQ FILES)

- INPUT FILES : R1, R2 FASTQ FILES
- TOOL : FASTP
- TIME TAKEN: (1 HRS FOR 16 SAMPLES)
- DATA SIZE : 120 GB (FINAL OUTPUT)

FASTQC RESULT:

ADAPTER TRIMMING OF FASTQ FILES

BEFORE AND AFTER FASTP READS COMPARISSION:

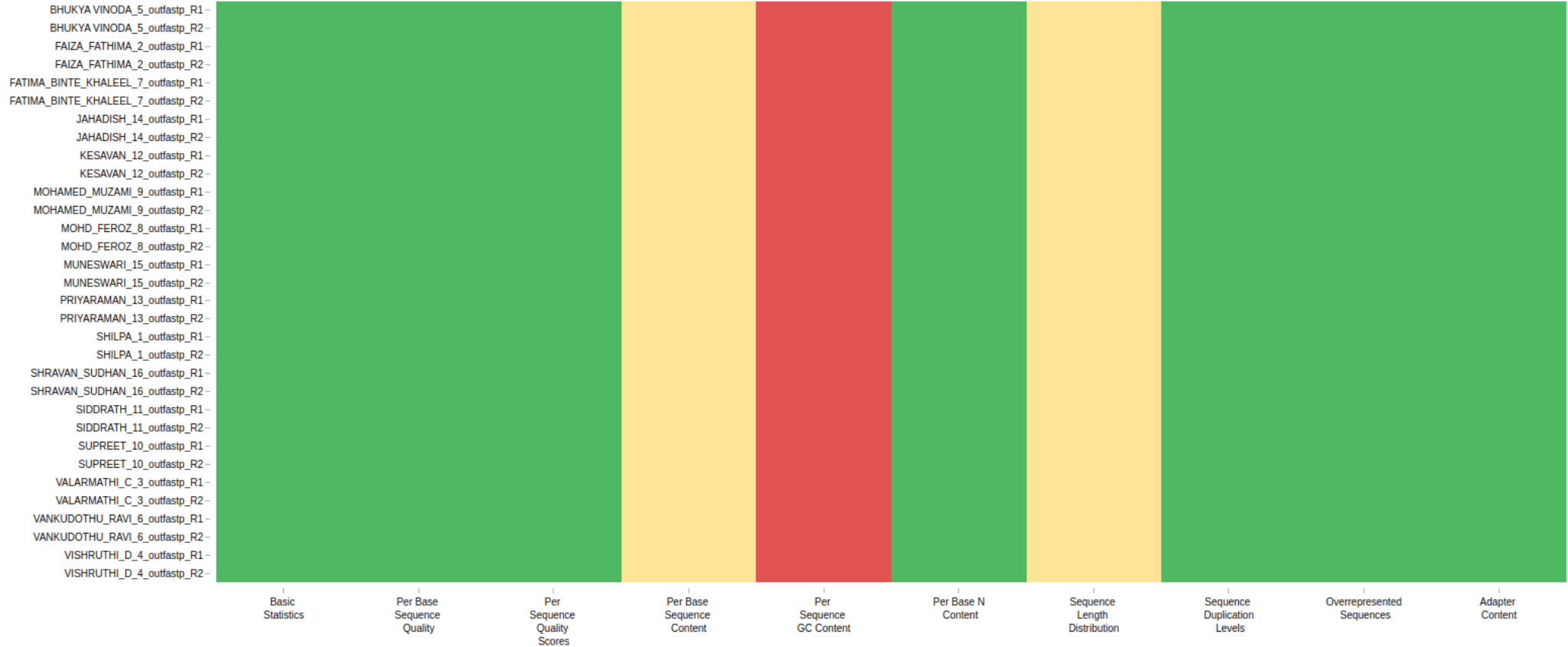
GENEMIND - BATCH -2 MAY							
BEFORE FASTP				AFTER FASTP			
Sample_ID	Sample_Name	NUMBER OF READS	SIZE OF READS	TOTAL DATA GENERATED	NUMBER OF READS	SIZE OF READS	TOTAL DATA GENERATED
1	SHILPA_1 - TWIST	84.372866 M	150bp	12.655930 G	84.133070 M	138bp	11.615106 G
2	FAIZA FATHIMA_2 - TWIST	87.742718 M	150bp	13.161408 G	87.493398 M	140bp	12.282293 G
3	VALARMATHI_C_3 - TWIST	89.831278 M	150bp	13.474692 G	89.566110 M	142bp	12.787434 G
4	VISHRUTHI_D_4 - TWIST	89.742578 M	150bp	13.461387 G	89.471114 M	144bp	12.969037 G
5	BHUKYA VINODA_5 - TWIST	63.396550 M	150bp	9.509483 G	63.276932 M	114bp	7.236525 G
6	VANKUDOTHU RAVI KIRAN_6 - TWIST	113.758242 M	150bp	17.063736 G	113.439252 M	132bp	15.055779 G
7	FATIMA BINTE KHALEEL_7 - TWIST	88.007242 M	150bp	13.201086 G	87.797598 M	120bp	10.560397 G
8	MOHD FEROZ_8 - TWIST	101.246270 M	150bp	15.186941 G	100.973756 M	129bp	13.116011 G
9	MOHAMED MUZAMI_9 - Nonaccus Exome	73.575836 M	150bp	11.036375 G	73.358920 M	121bp	8.884305 G
10	SUPREET_10 - Nonaccus Exome	67.163006 M	150bp	10.074451 G	67.006624 M	111bp	7.465811 G
11	SIDDRATH_11 - Nonaccus Exome	72.861586 M	150bp	10.929238 G	72.651454 M	122bp	8.927712 G
12	KESAVAN_12 - Nonaccus Exome	83.448974 M	150bp	12.517346 G	83.187120 M	126bp	10.501458 G
13	PRIYARAMAN_13 - Nonaccus Exome	74.290968 M	150bp	11.143645 G	74.056412 M	127bp	9.445938 G
14	JAHADISH_14 - Nonaccus Exome	80.498520 M	150bp	12.074778 G	80.240496 M	130bp	10.511361 G
15	MUNESWARI_15 - Nonaccus Exome	72.142778 M	150bp	10.821417 G	71.968782 M	112bp	8.063646 G
16	SHRAVAN SUDHAN_16 - Nonaccus Exome	72.920910 M	150bp	10.938137 G	72.730278 M	120bp	8.764775 G

STEP : 3 – MULTIQC ANALYSIS AFTER TRIMMING

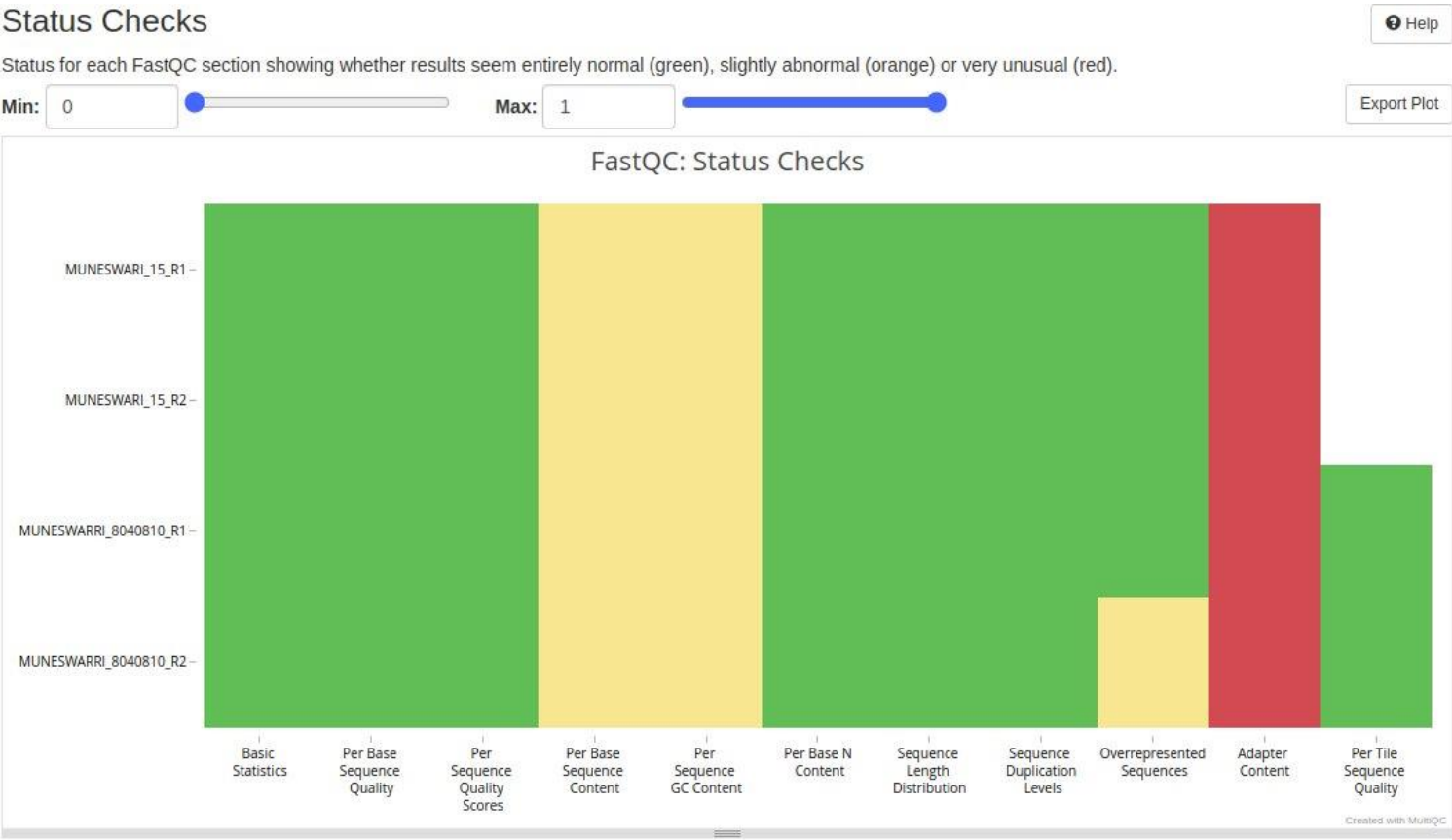
- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : FASTQC, MULTIQC
- TIME TAKEN: (2 HRS FOR 16 SAMPLES)
- DATA SIZE : 15 MB (FINAL OUTPUT)

FASTQC RESULT AFTER TRIMMING:

FastQC: Status Checks



MEDGENOME AND GENEMIND COMPARSSION FASTQC RESULT BEFORE FASTP:



MEDGENOME AND GENEMIND COMPARSSION FASTQC RESULT AFTER FASTP:

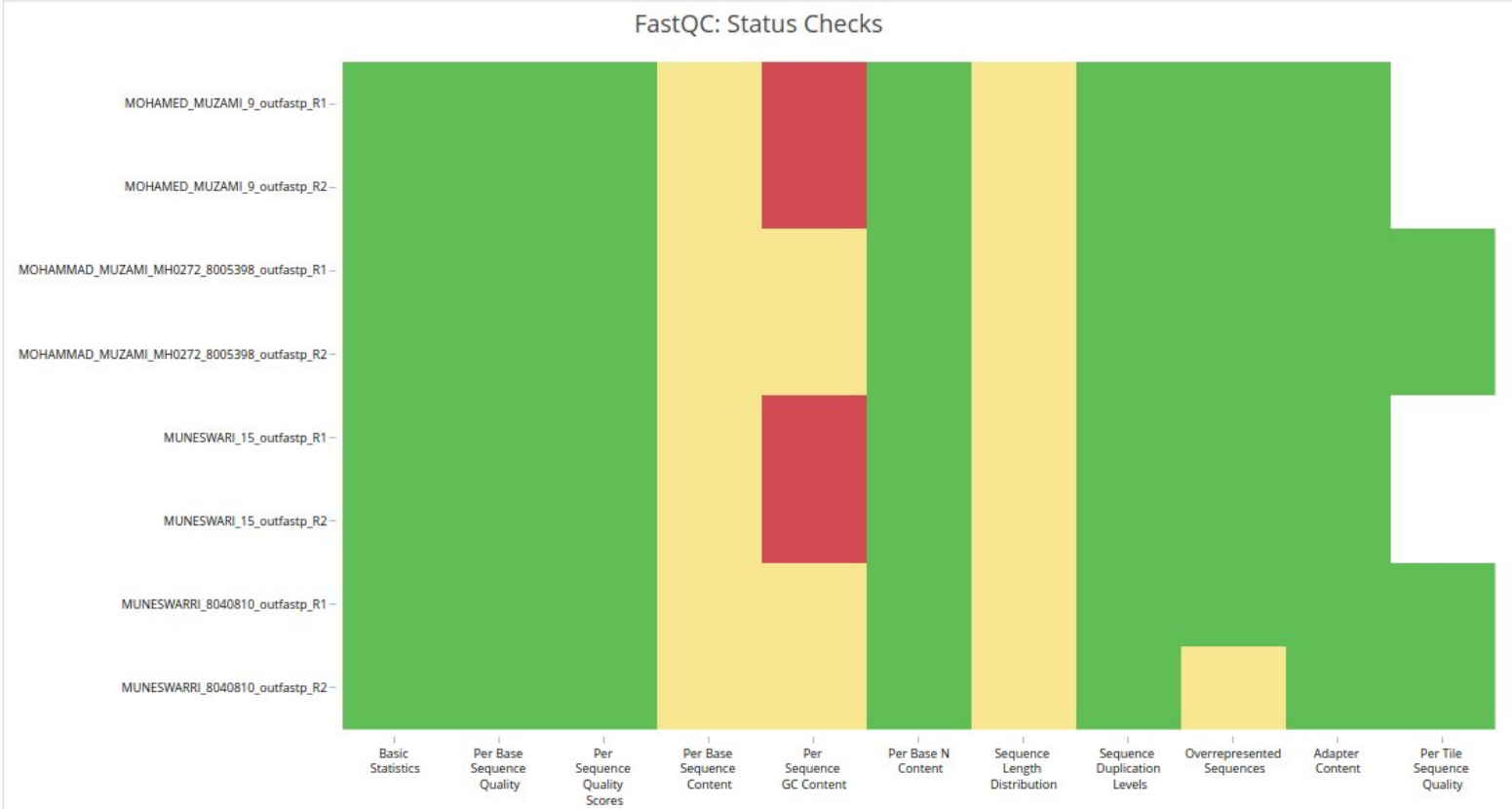
Status Checks

Help

Status for each FastQC section showing whether results seem entirely normal (green), slightly abnormal (orange) or very unusual (red).

Min: Max:

Export Plot



STEP 4 : HS METRICS ANALYSIS :

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : SENTIEON
- TIME TAKEN: (4 HRS FOR 16 SAMPLES)
- DATA SIZE : 100 GB (FINAL OUTPUT)

RESULT: HS METRICS (TWIST BED):

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000								
BAIT_SET		BAIT						
		TWIST BED						
BAIT_SET	SHILPA_1	FAIZA_2	VALARMATHI_C_3	VISHRUTHI_D_4	BHUKYA_5	VANKUDOTHU_6	FATIMA_7	MOHD_8
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815
TARGET_TERRITORY	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262
BAIT_DESIGN_EFFICIENCY	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969
TOTAL_READS	84133070	87493398	89566110	89471114	63276932	113439252	87797598	100973756
PF_READS	84133070	87493398	89566110	89471114	63276932	113439252	87797598	100973756
PF_UNIQUE_READS	84133070	87493398	89566110	89471114	63276932	113439252	87797598	100973756
PCT_PF_READS	1	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	84126716	87487816	89560341	89466343	63271189	113429176	87785517	100963955
PCT_PF_UQ_READS_ALIGNED	0.999924	0.999936	0.999936	0.999947	0.999909	0.999911	0.999862	0.999903
PF_BASES_ALIGNED	11542857727	12226529855	12752065054	12938605052	7149911230	14952041756	10424745076	13027127729
PF_UQ_BASES_ALIGNED	11542857727	12226529855	12752065054	12938605052	7149911230	14952041756	10424745076	13027127729
ON_BAIT_BASES	7821691493	8159304157	8083600818	8058425408	5638214126	11041963449	8106357589	9719172390
NEAR_BAIT_BASES	2805912273	3139053584	3580036572	3857784253	1075753537	3077986574	1722508474	2506868464
OFF_BAIT_BASES	915253961	928172114	1088427664	1022395391	435943567	832091733	595879013	801086875
ON_TARGET_BASES	4763995637	4874348071	4847950494	4796409044	4197601514	7094497538	5765992002	6489099064
PCT_SELECTED_BASES	0.920708	0.924085	0.914647	0.920981	0.939028	0.944349	0.94284	0.938506
PCT_OFF_BAIT	0.0792918	0.0759146	0.0853531	0.079019	0.0609719	0.0556507	0.0571601	0.0614937
ON_BAIT_VS_SELECTED	0.735979	0.722167	0.69306	0.676257	0.839774	0.782012	0.82475	0.794957
MEAN_BAIT_COVERAGE	180.849	188.655	186.905	186.323	130.364	255.307	187.431	224.722
MEAN_TARGET_COVERAGE	130.67	133.697	132.973	131.559	115.134	194.592	158.153	177.987
PCT_USABLE_BASES_ON_BAIT	0.673407	0.664314	0.632152	0.621359	0.779133	0.733404	0.767619	0.741016
PCT_USABLE_BASES_ON_TARGET	0.410155	0.39686	0.379118	0.369835	0.580058	0.471214	0.546001	0.494746
FOLD_ENRICHMENT	50.4082	49.6436	47.1561	46.3315	58.6617	54.9363	57.8461	55.5002
ZERO_CVG_TARGETS_PCT	0.00830451	0.00831156	0.00742405	0.0073994	0.0111326	0.00842426	0.0105198	0.00884336
PCT_EXC_DUPE	0	0	0	0	0	0	0	0
PCT_EXC_MAPQ	0.0340817	0.0332588	0.0323774	0.029245	0.0396927	0.0353758	0.0396479	0.0374629
PCT_EXC_BASEQ	0.0264073	0.0254977	0.0244125	0.0244773	0.0307795	0.0305559	0.0350009	0.0295624
PCT_EXC_OVERLAP	0.395892	0.379354	0.332838	0.305708	0.467413	0.430119	0.461204	0.443141
PCT_EXC_OFF_TARGET	0.747015	0.737715	0.747427	0.749134	0.876249	0.768622	0.844486	0.791828
FOLD_80_BASE_PENALTY	1.71934	1.78262	1.70478	1.80218	1.69315	1.55674	1.63045	1.56129
PCT_TARGET_BASES_1X	0.98838	0.988481	0.989649	0.989838	0.985573	0.988834	0.986292	0.98822
PCT_TARGET_BASES_2X	0.987588	0.987771	0.988948	0.989076	0.984964	0.988218	0.985724	0.98759
PCT_TARGET_BASES_10X	0.984924	0.985082	0.986294	0.986122	0.982101	0.986179	0.983438	0.985403
PCT_TARGET_BASES_20X	0.981383	0.981266	0.982882	0.981853	0.977143	0.984456	0.981078	0.983564
PCT_TARGET_BASES_30X	0.974292	0.97365	0.976248	0.972557	0.965852	0.982346	0.977366	0.981095
PCT_TARGET_BASES_40X	0.959685	0.957642	0.962531	0.953479	0.942902	0.979119	0.970695	0.977039
PCT_TARGET_BASES_50X	0.932731	0.928638	0.937416	0.920969	0.904055	0.97395	0.959115	0.970268
PCT_TARGET_BASES_100X	0.63678	0.633047	0.650508	0.622929	0.549492	0.88864	0.780626	0.857937
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0
HS_PENALTY_10X	0	0	0	0	0	0	0	0
HS_PENALTY_20X	0	0	0	0	0	0	0	0
HS_PENALTY_30X	0	0	0	0	0	0	0	0
HS_PENALTY_40X	0	0	0	0	0	0	0	0
HS_PENALTY_50X	0	0	0	0	0	0	0	0
HS_PENALTY_100X	0	0	0	0	0	0	0	0
AT_DROPOUT	0.0257279	0.0234139	0.0230409	0.0220716	0.236946	0.382733	0.323308	0.42835
GC_DROPOUT	15.1403	16.5183	15.0154	15.9895	8.29136	8.12574	7.69747	7.04269

HS METRICS (NONACCUS BED):

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000								
BAIT_SET	BAIT							
	NONACCUS BED							
BAIT_SET	MOHAMED_9	SUPREET_10	SIDDRATH_11	KESAVAN_12	PRIYARAMAN_13	JAHADISH_14	MUNESWARI_15	SHRAVAN_16
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	47015525	47015525	47015525	47015525	47015525	47015525	47015525	47015525
TARGET_TERRITORY	47015525	47015525	47015525	47015525	47015525	47015525	47015525	47015525
BAIT_DESIGN_EFFICIENCY	1	1	1	1	1	1	1	1
TOTAL_READS	73358920	67006624	72651454	83187120	74056412	80240496	71968782	72730278
PF_READS	73358920	67006624	72651454	83187120	74056412	80240496	71968782	72730278
PF_UNIQUE_READS	73358920	67006624	72651454	83187120	74056412	80240496	71968782	72730278
PCT_PF_READS	1	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	73347394	66997708	72644041	83177604	74049221	80233131	71960186	72721734
PCT_PF_UQ_READS_ALIGNED	0.999843	0.999867	0.999898	0.999886	0.999903	0.999908	0.999881	0.999883
PF_BASES_ALIGNED	8809900864	7382128285	8862564503	10425478537	9386310133	10462246184	7979645545	8694193753
PF_UQ_BASES_ALIGNED	8809900864	7382128285	8862564503	10425478537	9386310133	10462246184	7979645545	8694193753
ON_BAIT_BASES	6628965628	5738853467	6674330429	7766760748	6957604379	7638057725	6206452650	6586061914
NEAR_BAIT_BASES	1494107924	1102511864	1514234097	1861289803	1712061083	2004396637	1196484866	1453893630
OFF_BAIT_BASES	686827312	540762954	673999977	797427986	716644671	819791822	576708029	654238209
ON_TARGET_BASES	5195355242	4833869847	5236430312	5960119475	5279310270	5716886947	5183941236	5249377152
PCT_SELECTED_BASES	0.922039	0.926747	0.92395	0.923512	0.92365	0.921643	0.927728	0.92475
PCT_OFF_BAIT	0.0779608	0.073253	0.0760502	0.0764884	0.07635	0.0783572	0.0722724	0.07525
ON_BAIT_VS_SELECTED	0.816066	0.838846	0.815079	0.806681	0.802523	0.792128	0.838377	0.819166
MEAN_BAIT_COVERAGE	140.995	122.063	141.96	165.196	147.985	162.458	132.009	140.083
MEAN_TARGET_COVERAGE	110.503	102.814	111.377	126.769	112.289	121.596	110.26	111.652
PCT_USABLE_BASES_ON_BAIT	0.746143	0.768685	0.747597	0.739589	0.736571	0.726648	0.769683	0.751424
PCT_USABLE_BASES_ON_TARGET	0.584779	0.647467	0.586537	0.567552	0.558897	0.543877	0.642878	0.598918
FOLD_ENRICHMENT	51.491	53.1986	51.5353	50.9801	50.7249	49.9591	53.2251	51.8386
ZERO_CVG_TARGETS_PCT	0.0122873	0.0133052	0.0124505	0.0120568	0.0118215	0.0116583	0.0144144	0.0124025
PCT_EXC_DUPE	0	0	0	0	0	0	0	0
PCT_EXC_MAPQ	0.0601665	0.0590743	0.0573463	0.0568919	0.055024	0.0540716	0.0604771	0.0567165
PCT_EXC_BASEQ	0.0355991	0.0360505	0.0351868	0.0339742	0.035315	0.0307578	0.0363449	0.0355528
PCT_EXC_OVERLAP	0.43094	0.455836	0.435084	0.423834	0.420759	0.405582	0.45351	0.439423
PCT_EXC_OFF_TARGET	0.797984	0.854272	0.799806	0.783392	0.777951	0.763968	0.848842	0.812009
FOLD_80_BASE_PENALTY	1.78231	1.80376	1.76788	1.76068	1.75451	1.73708	1.80754	1.77225
PCT_TARGET_BASES_1X	0.986602	0.985139	0.986298	0.986983	0.987094	0.987282	0.984053	0.986325
PCT_TARGET_BASES_2X	0.985755	0.984336	0.985514	0.986171	0.986267	0.986484	0.983273	0.985495
PCT_TARGET_BASES_10X	0.982932	0.981347	0.982784	0.983687	0.983557	0.984026	0.980482	0.982736
PCT_TARGET_BASES_20X	0.978454	0.97433	0.978239	0.980693	0.979432	0.980995	0.974786	0.978216
PCT_TARGET_BASES_30X	0.965243	0.953871	0.964852	0.973143	0.967455	0.973072	0.95832	0.965364
PCT_TARGET_BASES_40X	0.935274	0.91259	0.934439	0.954789	0.939376	0.953355	0.924073	0.936251
PCT_TARGET_BASES_50X	0.884624	0.850103	0.883867	0.921123	0.89127	0.916921	0.870856	0.887354
PCT_TARGET_BASES_100X	0.49574	0.455674	0.502498	0.588138	0.509244	0.564341	0.505176	0.51031
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0
HS_PENALTY_10X	0	0	0	0	0	0	0	0
HS_PENALTY_20X	0	0	0	0	0	0	0	0
HS_PENALTY_30X	0	0	0	0	0	0	0	0
HS_PENALTY_40X	0	0	0	0	0	0	0	0
HS_PENALTY_50X	0	0	0	0	0	0	0	0
HS_PENALTY_100X	0	0	0	0	0	0	0	0
AT_DROPOUT	0.857427	0.964022	0.721797	0.745662	0.512678	0.542603	0.810977	0.694929
GC_DROPOUT	3.09792	2.3393	3.8169	3.94105	4.0904	3.96681	3.36929	2.9653

HS METRICS (TWIST AND NONACCUS BED):

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000																		
BAIT_SET	TWIST BED								NONACCUS BED									
	SHILPA_1	FAIZA_2	VALARMATHI_C_3	VISHRUTHI_D_4	BHUKYA_5	VANKUDOTHU_6	FATIMA_7	MOHD_8	MOHAMED_9	SUPREET_10	SIDDDRATH_11	KESAVAN_12	PRIYARAMAN_13	JAHADISH_14	MUNESWARI_15	SHRAVAN_16		
BAIT_SET	SHILPA_1	FAIZA_2	VALARMATHI_C_3	VISHRUTHI_D_4	BHUKYA_5	VANKUDOTHU_6	FATIMA_7	MOHD_8	MOHAMED_9	SUPREET_10	SIDDDRATH_11	KESAVAN_12	PRIYARAMAN_13	JAHADISH_14	MUNESWARI_15	SHRAVAN_16		
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815	47015525	47015525	47015525	47015525	47015525	47015525	47015525	47015525	47015525	47015525
TARGET_TERRITORY	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262	47015525	47015525	47015525	47015525	47015525	47015525	47015525	47015525	47015525	47015525
BAIT_DESIGN_EFFICIENCY	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	1	1	1	1	1	1	1	1	1	1
TOTAL_READS	84133070	87493398	89566110	89471114	63276932	113439252	87797598	100973756	73358920	67006624	72651454	83187120	74056412	80240496	71968782	72730278		
PF_READS	84133070	87493398	89566110	89471114	63276932	113439252	87797598	100973756	73358920	67006624	72651454	83187120	74056412	80240496	71968782	72730278		
PF_UNIQUE_READS	84133070	87493398	89566110	89471114	63276932	113439252	87797598	100973756	73358920	67006624	72651454	83187120	74056412	80240496	71968782	72730278		
PCT_PF_READS	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	84126716	87487816	89560341	89466343	63271189	113429176	87785517	100963955	73347394	66997708	72644041	83177604	74049221	80233131	71960186	72721734		
PCT_PF_UQ_READS_ALIGNED	0.999924	0.999936	0.999936	0.999947	0.999909	0.999911	0.999862	0.999903	0.999843	0.999867	0.999898	0.999886	0.999903	0.999908	0.999881	0.999883		
PF_BASES_ALIGNED	11542857727	12226529855	12752065054	12938605052	7149911230	14952041756	10424745076	13027127729	8809900864	7382128285	8862564503	10425478537	9386310133	10462246184	7979645545	8694193753		
PF_UQ_BASES_ALIGNED	11542857727	12226529855	12752065054	12938605052	7149911230	14952041756	10424745076	13027127729	8809900864	7382128285	8862564503	10425478537	9386310133	10462246184	7979645545	8694193753		
ON_BAIT_BASES	7821691493	8159304157	8083600818	8058425408	5638214126	11041963449	8106357589	9719172390	6628965828	5738853467	6674330429	7766760748	6957604379	7638057725	6206452650	6586061914		
NEAR_BAIT_BASES	2805912273	3139053584	3580036572	3857784253	1075753537	3077986574	1722508474	2506868464	1494107924	1102511864	1514234097	1861289803	1712061083	2004396637	1196484866	1453893630		
OFF_BAIT_BASES	915253961	928172114	1088427664	1022395391	435943567	832091733	595879013	801086875	686827312	540762954	673999977	797427986	716644671	819791822	576708029	654238209		
ON_TARGET_BASES	4763995637	4874348071	4847950494	4796409044	4197601514	7094497538	5765992002	6489099064	5195355242	4833869847	5236430312	5960119475	5279310270	5716886947	5183941236	5249377152		
PCT_SELECTED_BASES	0.920708	0.924085	0.914647	0.920981	0.939028	0.944349	0.94284	0.938506	0.922039	0.926747	0.92395	0.923512	0.92365	0.921643	0.927728	0.92475		
PCT_OFF_BAIT	0.0792918	0.0759146	0.0853531	0.079019	0.0609719	0.0556507	0.0571601	0.0614937	0.0779608	0.073253	0.0760502	0.0764884	0.07635	0.0783572	0.0722724	0.07525		
ON_BAIT_VS_SELECTED	0.735979	0.722167	0.69306	0.676257	0.839774	0.782012	0.82475	0.794957	0.838846	0.815079	0.806681	0.802523	0.792128	0.838377	0.819166			
MEAN_BAIT_COVERAGE	180.849	188.655	186.905	186.323	130.364	255.307	187.431	224.722	140.995	122.063	141.96	165.196	147.985	162.458	132.009	140.083		
MEAN_TARGET_COVERAGE	130.67	133.697	132.973	131.559	115.134	194.592	158.153	177.987	110.503	102.814	111.377	126.769	112.289	121.596	110.26	111.652		
PCT_USABLE_BASES_ON_BAIT	0.673407	0.664314	0.632152	0.621359	0.779133	0.733404	0.767619	0.741016	0.768685	0.736571	0.747597	0.739589	0.736571	0.726648	0.769683	0.751424		
PCT_USABLE_BASES_ON_TARGET	0.410155	0.39686	0.379118	0.369835	0.580058	0.471214	0.546001	0.494746	0.584779	0.647467	0.586537	0.567552	0.558897	0.543877	0.642878	0.598918		
FOLD_ENRICHMENT	50.4082	49.6436	47.1561	46.3315	58.0617	54.9363	57.8461	55.5002	51.4991	53.1986	51.5353	50.9801	50.7249	49.9591	53.2251	51.8386		
ZERO_CVG_TARGETS_PCT	0.00830451	0.00831156	0.00742405	0.0073994	0.0111326	0.00842426	0.0105198	0.00884336	0.0122873	0.0133052	0.0124505	0.0120568	0.0118215	0.0116583	0.0144144	0.0124025		
PCT_EXC_DUPE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
PCT_EXC_MAPQ	0.0340817	0.0332588	0.0323774	0.029245	0.0396927	0.0353758	0.0396479	0.0374629	0.0601665	0.0590743	0.0573463	0.0568919	0.055024	0.0540716	0.0604771	0.0567165		
PCT_EXC_BASEQ	0.0264073	0.0254977	0.0244125	0.0244773	0.0307795	0.0305559	0.0350009	0.0295624	0.0355991	0.0360505	0.0351868	0.0339742	0.035315	0.0307578	0.0363449	0.0355528		
PCT_EXC_OVERLAP	0.395892	0.379354	0.332838	0.305708	0.467413	0.430119	0.461204	0.443141	0.43094	0.455836	0.435084	0.423834	0.420759	0.405582	0.45351	0.439423		
PCT_EXC_OFF_TARGET	0.747015	0.737715	0.747427	0.749134	0.876249	0.768622	0.844486	0.791828	0.797984	0.854272	0.799806	0.783392	0.777951	0.763968	0.848842	0.812009		
FOLD_80_BASE_PENALTY	1.71934	1.78262	1.70478	1.80218	1.69315	1.55674	1.63045	1.56129	1.78231	1.80376	1.76788	1.76068	1.75451	1.73708	1.80754	1.77225		
PCT_TARGET_BASES_1X	0.98838	0.988481	0.989649	0.989838	0.985573	0.988834	0.986292	0.98822	0.986602	0.985139	0.986298	0.986983	0.987094	0.987822	0.984053	0.986325		
PCT_TARGET_BASES_2X	0.987588	0.987771	0.988948	0.989076	0.984964	0.988218	0.985724	0.98759	0.985755	0.984336	0.985514	0.986171	0.986267	0.986484	0.983273	0.985495		
PCT_TARGET_BASES_10X	0.984924	0.985082	0.986294	0.986122	0.982101	0.986179	0.983438	0.985403	0.982932	0.981347	0.982784	0.983687	0.984026	0.984026	0.980482	0.982736		
PCT_TARGET_BASES_20X	0.981383	0.981266	0.982882	0.981853	0.977143	0.984456	0.981078	0.983564	0.978454	0.97433	0.978239	0.980693	0.979432	0.980995	0.974786	0.978216		
PCT_TARGET_BASES_30X	0.974292	0.97365	0.976248	0.972557	0.965852	0.982346	0.977366	0.981095	0.965243	0.953871	0.964852	0.973143	0.967455	0.973072	0.95832	0.965364		
PCT_TARGET_BASES_40X	0.959685	0.957642	0.962531	0.953479	0.942902	0.979119	0.970695	0.977039	0.935274	0.91259	0.934439	0.954789	0.939376	0.953355	0.924073	0.936251		
PCT_TARGET_BASES_50X	0.932731	0.928638	0.937416	0.920969	0.904055	0.97395	0.911213	0.970268	0.884624	0.850103	0.883867	0.921123	0.89127	0.916921	0.870856	0.887354		
PCT_TARGET_BASES_100X	0.63678	0.633047	0.650508	0.622929	0.549492	0.88864	0.780626	0.857937	0.49574	0.455674	0.502498	0.588138	0.509244	0.564341	0.505176	0.51031		
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
HS_PENALTY_10X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
HS_PENALTY_20X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
HS_PENALTY_30X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
HS_PENALTY_40X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
HS_PENALTY_50X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
HS_PENALTY_100X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
AT_DROPOUT	0.0257279	0.0234139	0.0230409	0.0220716	0.236946	0.382733	0.323308	0.42835	0.857427	0.964022	0.721797	0.745662	0.512678	0.542603	0.810977	0.694929		
GC_DROPOUT	15.1403	16.5183	15.0154	15.9895	8.29136	8.12574	7.69747	7.04269	3.09792	2.3393	3.8169	3.94105	4.0904	3.96681	3.36929	2.9653		

STEP 5 : SECONDARY ANALYSIS (BAM AND VCF GENERATION):

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : SENTIEON
- TIME TAKEN: (6 HRS FOR 16 SAMPLES)
- DATA SIZE : 130 GB (FINAL OUTPUT)

SECONDARY ANALYSIS RESULT:

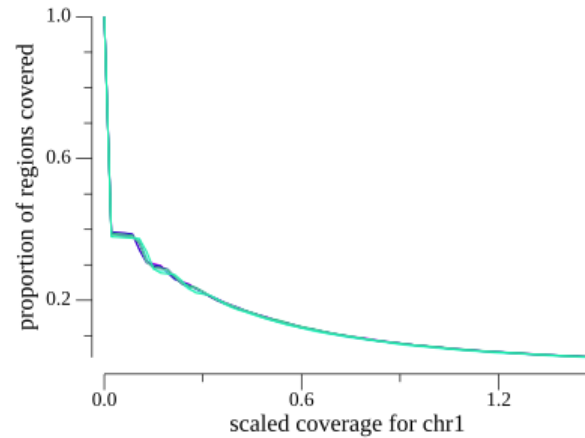
VCF AND BAM FILES

STEP 6 : INDEXCOVERAGE

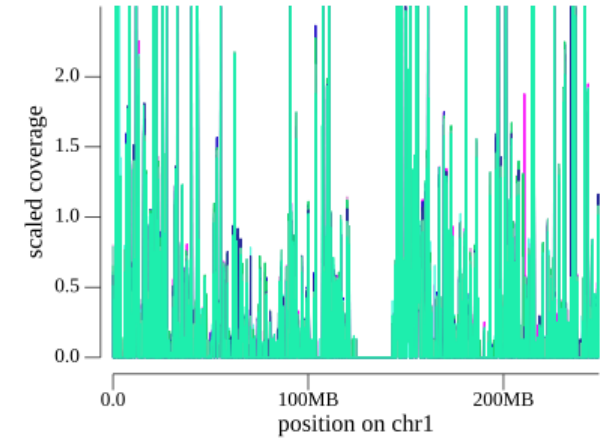
- INPUT FILES : BAM FILES
- TOOL : INDEXCOV
- TIME TAKEN: (1 HRS)
- DATA SIZE : 100 MB (FINAL OUTPUT)

INDEXCOVERAGE NONACCUS BAM (8 SAMPLES):

Click each plot for an interactive view.

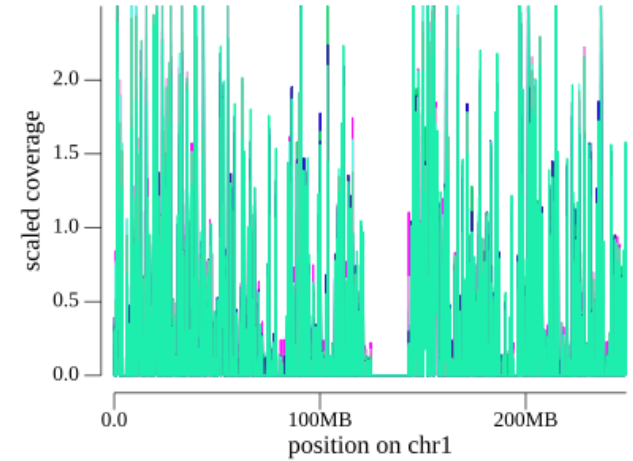
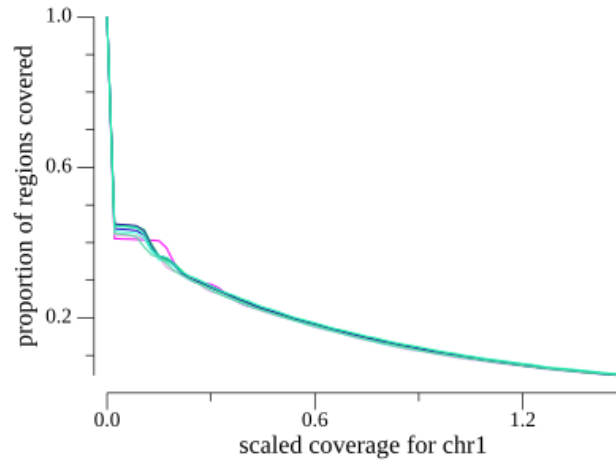


Click each plot for an interactive view.



ALL CHROMOSOME COVERAGE STATISTICS AVAILABLE IN HTML LINK

INDEXCOVERAGE TWIST BAM (8 SAMPLES) :



ALL CHROMOSOME COVERAGE STATISTICS AVAILABLE IN [HTML LINK](#)

STEP 7 : VS CLINICAL SAMPLE AND COVERAGE STASTICS:

- INPUT FILES : VCF, BAM, BED FILES
- TOOL : VARSEQ
- TIME TAKEN: (2 DAYS 16 SAMPLES)
- DATA SIZE : 8 GB (FINAL OUTPUT)

RESULT: SAMPLE STASTICS (TWIST BED 8 SAMPLES):

SAM PLE STASTICS								
	SHILPA_1	FAIZA_FATHIM_A_2	VALARM_ATHI_C_3	VISHRUTHI_D_4	BHUKYA_VINODA_5	VANKUDOTHU_RAVI_KIRAN_6	FATIM_A_BINTE_KHALEEL_7	M_OHD_FEROZ_8
	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)
Transition Count	137153	133059	169196	155273	83419	101072	84456	113666
Transversion Count	66897	66162	86653	80052	37737	46909	37963	53165
Ti/ Tv	2.050211519	2.011109096	1.952569444	1.939651726	2.210536079	2.154639835	2.224692464	2.137985517
Coding Transition Count	20482	20868	21134	21509	19269	20036	19407	20134
Coding Transversion Count	7441	7770	7968	8303	6770	7202	6876	7107
Coding Ti/ Tv	2.752587018	2.685714286	2.652359438	2.590509454	2.846233383	2.782004999	2.822425829	2.832981567
Singleton Count	237568	232169	295872	271174	141728	173475	143482	195362
Het Count	92077	98344	116809	121313	57462	78483	63433	73916
Het Ratio	0.387581661	0.42358799	0.394795722	0.44736221	0.405438587	0.452416775	0.442097267	0.37835403
Hom Count	145491	133825	179063	149861	84266	94992	80049	121446
Hom Ratio	0.613567642	0.577730866	0.60637862	0.553750707	0.595662562	0.548950261	0.559200553	0.623017257
Hemi Count	0	0	0	0	0	0	0	0
Hemi Ratio	0	0	0	0	0	0	0	0
SNV Het Count	77879	83017	99704	104188	48094	65351	52715	61565
SNV Hom Count	125988	116009	155920	130936	72969	82483	69614	105119
SNV Het/Hom Ratio	0.618146173	0.715608272	0.639456131	0.795716992	0.65910181	0.792296594	0.757247105	0.585669574
Indel Het Count	13308	14267	15959	16035	8844	12268	10052	11491
Indel Hom Count	19503	17816	23143	18925	11297	12509	10435	16327
Indel Het/Hom Ratio	0.682356561	0.800797036	0.689582163	0.847291942	0.782862707	0.980733872	0.963296598	0.703803516
Var Count	237568	232169	295872	271174	141728	173475	143482	195362
Coding Var Count	28895	29689	30134	30845	26989	28275	27266	28258
Ref Count	0	0	0	0	0	0	0	0
Ref Ratio	0	0	0	0	0	0	0	0
Called Genotypes	237568	232169	295872	271174	141728	173475	143482	195362
Call Rate	1	1	1	1	1	1	1	1
SNP Count	203989	199156	255774	235258	121125	147932	122389	166782
M NP Count	0	0	0	0	0	0	0	0
Ins Count	13524	13241	16219	14742	8349	10352	8552	11728
Del Count	19782	19454	23529	20833	12099	14925	12338	16585
DelIns Count	0	0	0	0	0	0	0	0
Complex Count	273	318	350	341	155	266	203	267
Chrom X Het Ratio	0.311126257	0.373345588	0.337248797	0.202085402	0.325757576	0.173895742	0.377641654	0.136125654
Inferred Gender	Female	Female	Female	Male	Female	Male	Female	Male
M ultiallelic Site Count	445	530	573	545	262	432	333	430
M ultiallelic Het Count	445	530	573	545	262	432	333	430
Coding M ultiallelic Site Count	30	30	30	28	28	34	24	37
Coding M ultiallelic Het Count	30	30	30	28	28	34	24	37
Q20 Values	10.898937 G (93.834160%)	11.530393 G (93.878173%)	12.008802 G (93.910967%)	12.163517 G (93.788901%)	6.858832 G (94.780748%)	14.105025 G (93.685119%)	9.897587 G (93.723631%)	12.348193 G (94.145949%)
Q30 Values	9.749109 G (83.934736%)	10.313082 G (83.967072%)	10.752907 G (84.089644%)	10.865732 G (83.782110%)	6.211903 G (85.840971%)	12.592657 G (83.640025%)	8.835705 G (83.668307%)	11.102454 G (84.648099%)

RESULT: SAMPLE STASTICS (NONACCUS BED 8 SAMPLES):

	SAMPLE STASTICS																							
	MOHAMED_MUZAM_I_9				SUPREET_10				SIDDRATH_11				KESAVAN_12				PRIYARAMAN_13				JAHADISH_14			
	NONACCUS BED	M_EDGENOM E BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	M_EDGENOM E BED	NONACCUS BED	KAPPA BED
Transition Count	80551	202666	70139	92206	79598	93964	82839	94134	83338	105816	88633	87747	84235	32373	70589	119757	78496	36623	46613	101934				
Transversion Count	37783	97525	32193	42050	37195	42050	30294	42524	36330	47799	40413	42305	37799	42305	32373	70589	119757	78496	36623	46613				
Ti/Tv	2.131937844	2.078092797	2.178703445	2.192770511	2.14001882	2.205313556	2.108721108	2.213667576	2.102851375	2.213770163	2.074151961	2.183180412	2.180489914	2.180742639	2.183180412	2.180742639	2.183180412	2.180742639	2.183180412	2.180742639				
Coding Transition Count	19464	22244	18663	21251	19374	21510	18930	20813	19572	21752	18746	20225	18341	19872	19342	21527	7016	8209	21527	7016				
Coding Transversion Count	7028	8422	6957	8146	6944	8152	6763	7808	7060	8293	7038	7720	6521	7306	6521	7306	6521	7306	6521	7306				
Coding Ti/Tv	2.768493465	2.641177867	2.803515097	2.608765038	2.790034562	2.63861629	2.799053674	2.665959385	2.72223796	2.622935005	2.663540779	2.619818653	2.812605429	2.7199562	2.7199562	2.7199562	2.812605429	2.7199562	2.812605429	2.7199562				
Singleton Count	137723	349577	119058	154214	136022	156684	142852	157691	142320	176799	152054	148066	148066	152054	148066	152054	148066	152054	148066	152054				
Het Count	70931	163010	60603	70871	79484	68272	72790	73967	85435	73967	85435	73967	85435	73967	85435	73967	85435	73967	85435	73967				
Het Ratio	0.515026539	0.466306422	0.509020813	0.520640149	0.521026001	0.507288555	0.477921205	0.461559895	0.52667039	0.483554771	0.486583714	0.464576608	0.490026204	0.486128264	0.519506743	0.498392771	0.486128264	0.519506743	0.498392771	0.486128264				
Hom Count	66792	186567	58455	77200	65151	77200	74580	84901	67795	91307	78067	79278	61090	101756	85882	101756	85882	101756	85882	101756				
Hom Ratio	0.486357778	0.535063417	0.492233906	0.480007272	0.480337079	0.493936715	0.523758804	0.539351896	0.474688419	0.517331838	0.515116924	0.536172976	0.511226223	0.503030363	0.481962018	0.502454361	0.503030363	0.481962018	0.502454361	0.503030363				
Hemi Count	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Hemi Ratio	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
SNV Het Count	59817	136727	51113	69559	59705	68846	56825	62237	63640	73700	61676	59426	49269	85881	58836	73716	85881	58836	73716	85881				
SNV Hom Count	58362	163029	51042	64577	56860	67567	65070	74283	59122	79735	68112	68479	53483	74696	56291	74696	56291	74696	56291	74696				
SNV Het/Hom Ratio	1.025106252	0.83866674	0.83866674	1.003391011	1.077148211	1.050303174	1.028290359	0.837836382	1.074418254	0.924311783	0.905506574	0.885308798	0.921226808	0.964489899	1.04511149	0.986880154	0.964489899	1.04511149	0.986880154	0.986880154				
Indel Het Count	10330	24467	8884	10315	10394	10114	10531	9997	10975	11186	11307	8948	8789	11137	11137	11137	8789	11137	11137	11137				
Indel Hom Count	8440	23538	9347	10394	8291	9633	10618	8673	9799	11572	9955	7979	7607	12714	8196	12714	7607	12714	8196	12714				
Indel Het/Hom Ratio	1.223933649	1.039468094	1.198435181	1.10356264	1.253648535	1.049932524	1.107360673	0.941514409	1.265421423	0.966643623	1.135861115	0.913151404	1.1553832	1.115456329	1.22852611	0.986797803	1.115456329	1.22852611	0.986797803	0.986797803				
Var Count	137723	349577	119058	154214	136022	156684	142852	157691	143230	176799	152054	148066	148066	152054	148066	152054	148066	152054	148066	152054				
Coding Var Count	27421	31897	26123	30760	27180	30994	26574	29925	27551	31339	26661	29094	25666	467	27253	30968	25666	467	27253	30968				
Ref Count	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Ref Ratio	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Called Genotypes	137723	349577	119058	154214	136022	156684	142852	157691	143230	176799	152054	148066	148066	152054	148066	152054	148066	152054	148066	152054				
Call Rate	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1				
SNP Count	118279	30046	102273	134216	116717	136519	122047	138612	122898	153555	129964	128999	128999	153555	128999	153555	128999	153555	128999	153555				
MNP Count	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Ins Count	8066	21911	6853	7884	9313	8701	9900	8546	10764	9900	8546	8847	12455	7781	10594	7781	12455	7781	10594	7781				
Del Count	11137	27114	9745	10581	11204	10724	11839	11032	12330	11558	12571	10122	9798	10930	12170	10930	9798	10930	12170	10930				
Del/Het Count	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Complex Count	241	506	187	100	217	128	265	147	228	150	285	98	203	240	244	147	203	240	244	147				
Chrom X Het Ratio	0.17963735	0.112922561	0.166130761	0.152777778	0.168610301	0.14913511	0.195335277	0.176420094	0.18825183	0.164108187	0.185473943	0.152602322	0.434858258	0.434858258	0.190169959	0.15785489	0.434858258	0.434858258	0.190169959	0.15785489				
Inferred Gender	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male	Male				
Multiallelic Site Count	392	908	303	208	386	262	458	278	410	303	402	207	349	289	349	289	349	289	349	289				
Multiallelic Het Count	392	908	303	208	386	262	458	278	410	303	402	207	349	289	349	289	349	289	349	289				
Coding Multiallelic Site Count	42	51	23	27	38	38	35	27	43	30	32	43	30	30	30	30	30	30	30	30				
Coding Multiallelic Het Count	42	51	23	27	38	38	35	27	43	30	32	43	30	30	30	30	30	30	30	30				
Q20 Values	8.340349 G (93.877337%)	10.361660 G (97.783918%)	7.046976 G (94.389967%)	4.479552 G (98.574592%)	8.391515 G (93.994024%)	5.387441 G (98.482209%)	9.871266 G (93.999007%)	5.630999 G (98.487317%)	8.853041 G (93.723266%)	6.768665 G (98.526834%)	9.915325 G (94.329604%)	3.914435 G (98.600616%)	7.605798 G (94.322078%)	9.057315 G (98.083435%)	8.243131 G (94.048407%)	5.531920 G (98.455847%)	7.605798 G (94.322078%)	9.057315 G (98.083435%)	8.243131 G (94.048407%)	5.531920 G (98.455847%)				
Q30 Values	7.477616 G (94.168581%)	9.331967 G (93.728669%)	6.362516 G (95.222037%)	4.355882 G (95.853172%)	7.529427 G (94.337705%)	5.232642 G (95.652403%)	8.864599 G (94.413030%)	5.468556 G (95.611176%)	7.904463 G (93.681082%)	6.576353 G (95.72779%)	8.950221 G (95.148072%)	3.807481 G (95.911842%)	6.853907 G (94.997621%)	8.923023 G (96.629589%)	7.401884 G (94.45084%)	5.369168 G (95.559272%)	6.853907 G (94.997621%)	8.923023 G (96.629589%)	7.401884 G (94.45084%)	5.369168 G (95.559272%)				

RESULT: COVERAGE STASTICS (TWIST BED):

COVERAGE STASTICS								
	SHILPA_1	FAIZA_FATHIM A_2	VALARM ATHI_ C_3	VISHRUTHI_ D_4	BHUKYA_VINODA_5	VANKUDOTHU_RAVI_KIRAN_6	FATIM A_BINTE_KHALEEL_7	MOHD_FEROZ_8
	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)	GENEM IND BED(TWIST v2.0.2)
Sample Mean Depth	169.235	176.004	175.107	176.748	101.334	202.308	133.8	175.416
Sample Mean Forward Depth	84.616	87.9968	87.551	88.3688	50.6651	101.15	66.8972	87.7066
Sample Mean Reverse Depth	84.6185	88.0069	87.556	88.3787	50.6683	101.158	66.9024	87.7094
Sample Mean Filtered Depth	3.95101	3.95509	3.79149	3.45368	3.16001	5.44976	4.28049	5.02534
Sample % 1x	98.8726	98.8864	98.9928	99.0135	98.5879	98.918	98.6656	98.8581
Sample % 5x	98.6804	98.6955	98.8142	98.8173	98.3983	98.7432	98.4882	98.6758
Sample % 20x	98.3349	98.3416	98.4624	98.4273	97.8096	98.5113	98.1609	98.4259
Sample % 50x	96.6547	96.5963	96.9163	96.3432	90.0922	97.8532	96.0006	97.5041
Sample % 100x	80.7403	80.8051	82.7874	80.4609	47.1835	92.5229	73.5112	88.7435
Sample % 500x	0.0508466	0.0662641	0.0743855	0.0659211	0.00539787	0.0832885	0.0144849	0.0412468
Sample Errors	?	?	?	?	?	?	?	?
Q20 Values	10.898937 G (93.834160%)	11.530393 G (93.878173%)	12.008802 G (93.910967%)	12.163517 G (93.788901%)	6.858832 G (94.780748%)	14.105025 G (93.685119%)	9.897587 G (93.723631%)	12.348193 G (94.145949%)
Q30 Values	9.749109 G (83.934736%)	10.313082 G (83.967072%)	10.752907 G (84.089644%)	10.865732 G (83.782110%)	6.211903 G (85.840971%)	12.592657 G (83.640025%)	8.835705 G (83.668307%)	11.102454 G (84.648099%)

RESULT: COVERAGE STASTICS (NONACCUS BED):

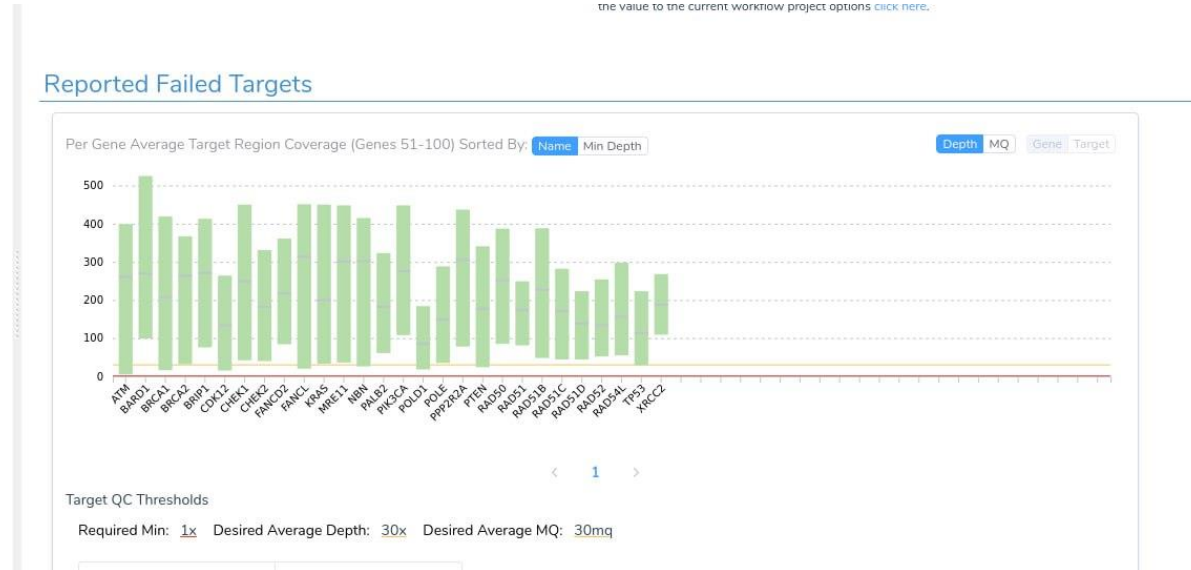
	COVERAGE STASTICS																			
	MOHAMED_MUZAMIL_9				SUPREET_10				SIDDRATH_11				KESAVAN_12				PRIYARAMAN_13			
	NONACCUS BED	M EDGENOM E BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	KAPPA BED	NONACCUS BED	M EDGENOM E BED	NONACCUS BED	KAPPA BED
Sample Mean Depth	128.353	112.049	111.36	48.6746	129.229	57.0914	149.777	60.0025	135.269	76.6264	148.639	41.823	119.664	140.854	128.207	60.3572				
Sample Mean Forward Depth	64.1772	56.0094	55.6796	24.3093	64.6141	28.5365	74.8887	29.9713	67.6351	38.2719	74.3213	20.884	59.8311	70.4238	64.103	30.1449				
Sample Mean Reverse Depth	64.1758	56.039	55.6807	24.3651	64.6152	28.5746	74.8879	30.0309	67.6339	38.3543	74.3178	20.9388	59.833	70.4303	64.1038	30.212				
Sample Mean Filtered Depth	4.87376	3.52188	4.15386	2.41985	4.73636	3.18715	5.49222	3.4302	4.8058	4.14181	5.05699	2.13501	4.77351	4.82152	4.6373	3.04733				
Sample % 1x	98.6923	98.1844	98.5447	98.0901	98.6594	98.0417	98.73	98.1037	98.7441	98.1016	98.7607	98.0675	98.4399	96.2547	98.6609	98.195				
Sample % 5x	98.4865	97.4208	98.354	97.6468	98.4658	97.6182	98.534	97.6663	98.5378	97.7333	98.5633	97.6075	98.2481	95.0243	98.4632	97.7271				
Sample % 20x	98.1022	96.1613	97.737	94.729	98.087	95.9517	98.2357	96.2703	98.1887	96.8988	98.2682	92.5874	97.7442	92.9437	98.0757	96.3987				
Sample % 50x	92.8931	88.3063	92.7817	45.8887	92.7899	62.371	96.3687	67.3437	93.9514	83.5589	95.5173	29.5862	90.1389	89.2746	92.7225	67.8971				
Sample % 100x	60.6895	55.4877	51.1392	1.37513	61.0314	3.60461	71.1934	4.79682	64.423	19.6978	71.0646	0.537923	56.4364	76.7213	60.8979	4.95321				
Sample % 500x	0.0571617	0.00959899	0.0296265	0.00666313	0.0456826	0.0102196	0.0661837	0.0085657	0.0532652	0.0149328	0.0684811	0.00437505	0.0345331	0.0242487	0.0461801	0.00826121				
Sample Errors	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?			
Q20 Values	8.340349 G (93.877337%)	10.361660 G (97.783918%)	7.046976 G (94.389967%)	4.479552 G (98.574592%)	8.391515 G (93.994024%)	5.387441 G (98.482209%)	9.871266 G (93.999007%)	5.630999 G (98.487317%)	8.853041 G (93.723266%)	6.768665 G (98.526834%)	9.915325 G (94.329604%)	3.914435 G (98.606061%)	7.605798 G (94.322078%)	9.057315 G (98.083435%)	8.243131 G (94.048074%)	5.531920 G (98.455847%)				
Q30 Values	7.477616 G (84.166581%)	9.931967 G (93.728865%)	6.362516 G (95.222037%)	4.355882 G (95.853172%)	7.529427 G (84.337705%)	5.232642 G (95.652483%)	8.864599 G (94.413030%)	5.486556 G (95.611176%)	7.904463 G (83.681082%)	6.576353 G (95.727479%)	8.950221 G (95.148072%)	3.807481 G (95.911842%)	6.853907 G (94.997621%)	8.923023 G (96.629158%)	7.401884 G (84.450364%)	5.3693168 G (95.559227%)				

CANCER GENE DEPTH COVERAGE: (VISHRUTHI_D_4)

GENE LIST :

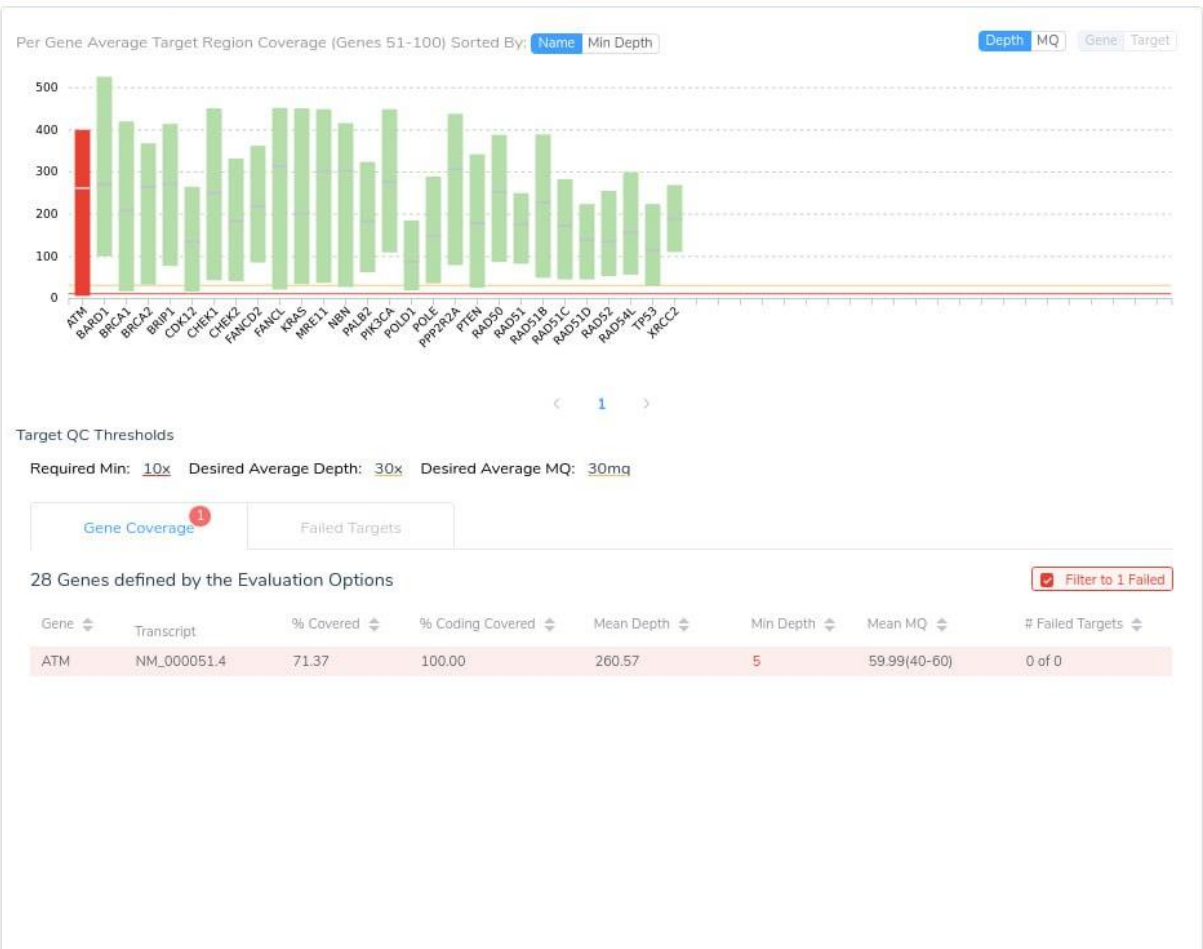
ATM, BARD1, BRCA1, BRCA2, BRIP1,
CDK12, CHEK2, FANCD2, MRE11, NBN,
PALB2, PPP2R2A, RAD51B, RAD54L,
TP53, CHEK1, FANCL, RAD50, RAD51,
RAD51C, RAD51D, RAD52, XRCC2,
KRAS, PIK3CA, POLD1, POLE, PTEN

1X :



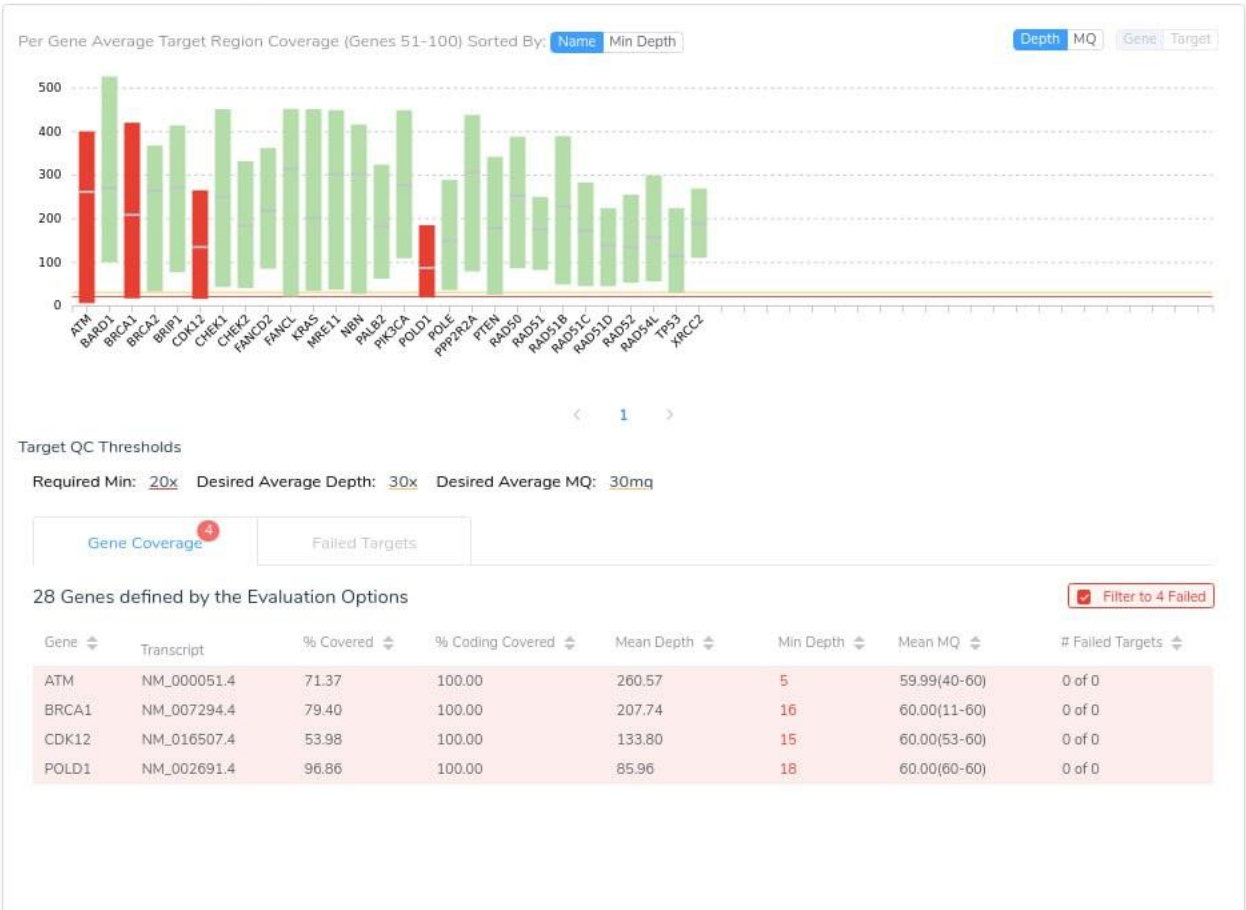
10X :

Reported Failed Targets



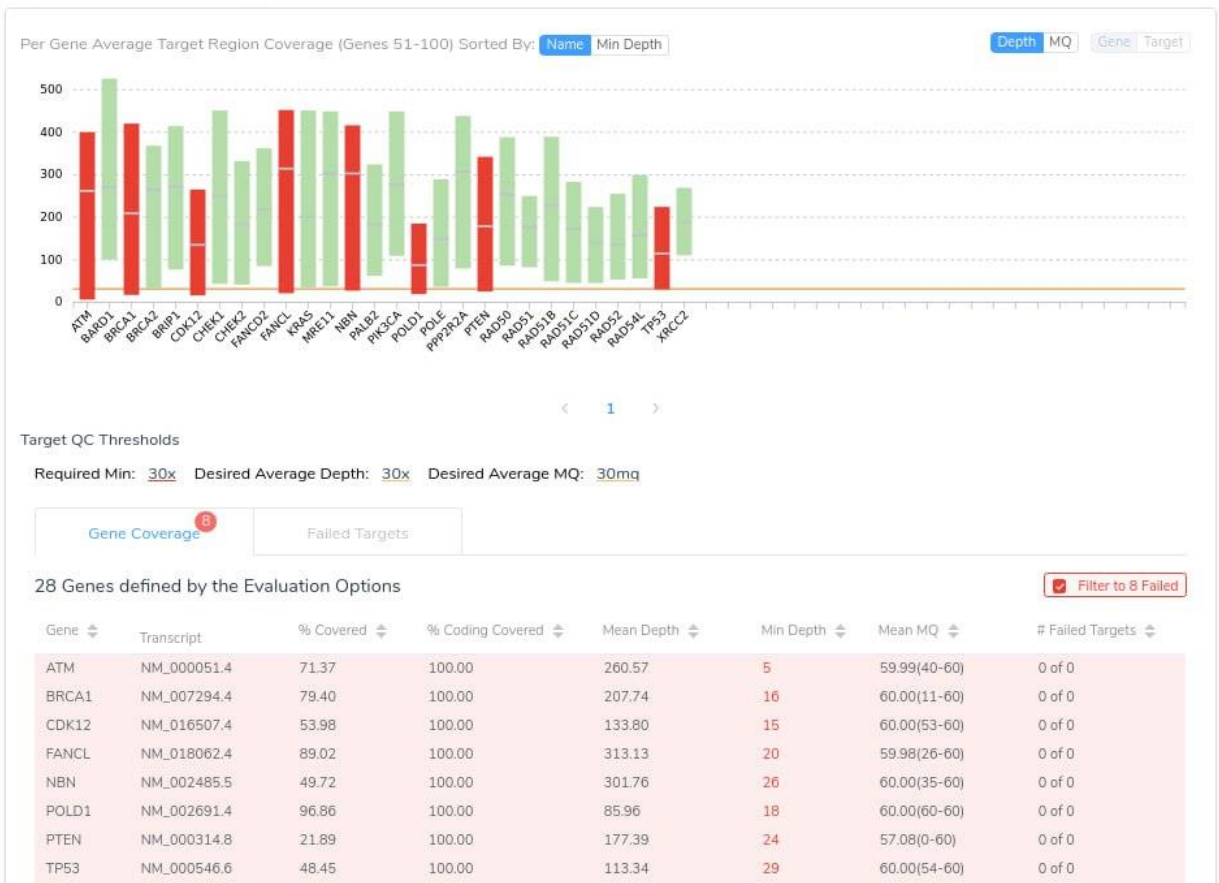
20X :

Reported Failed Targets



30X :

Reported Failed Targets



VARIANT DEPTH COMPARISSION :

SAMPLE_ID		SAMPLE NAME		DEPTH(REPORTED)	VARAINT ALLELE FREQUENCY	ALLELIC DEPTH	DEPTH (GENEMIND BED(TWIST)) - batch -2	VARAINT ALLELE FREQUENCY	ALLELIC DEPTH	GENE	LOCATION	VARIANT
1		SHILPA_1(GENE FEMINA)										
2		FAIZA_FATHIMA_2(GENE FEMINA)										
5		BHUKYA_VINODHA_5 – (ADVAT)										
6		VANKUDOTHU_RAVI_KIRAN_6 – (ADVAT)										
7		FATIMA_BHINTE_KHALEEL_7 – (ADVAT)										
8		FATIMA_BHINTE_KHALEEL_7 – (ADVAT)										
SAMPLE_ID		SAMPLE NAME		CNV TYPE(REPORTED)	SIZE	CHROMOSOME	CNV TYPE (GENEMIND BED(TWIST)) - batch -2	SIZE	CHROMOSOME			
3		VALARMATHI_C_3 -(CNV)		Gain	~90 Mb	CHROMOSOME 16		51.6 Mb	CHROMOSOME 16			
4		VISHRUTHI_D_4 – (CNV)		Loss	155 Mb	CHROMOSOME X		-				
SAMPLE_ID		SAMPLE NAME		DEPTH(REPORTED)	VARAINT ALLELE FREQUENCY	ALLELIC DEPTH	DEPTH (GENEMIND BED(NONACCUS)) - batch -2	VARAINT ALLELE FREQUENCY	ALLELIC DEPTH	GENE	LOCATION	VARIANT
9		Master. Mohammed Muzammi_9 - MEDGENOME BED		126x	1	0, 126	262x	1	0, 262	PKLR (-)	Exon 10	c.1529G>A (p. Arg510Gln)
10		Mr. Supreeth_10 - KAPPA BED		55x	0.436364	31, 24	178x	0.483146	92, 86	PAH	Exon 4	c.355C>T (p.Pro119Ser)
11		Master. Siddanth_11 - KAPPA BED		49x	0.489796	25, 24	131x	0.480916	68, 63	CHD8	Exon 17	c.3439G>C (p.Ala1147Pro)
				42x	0.47619	22, 20	100x	0.49	51, 49	AUTS2	Exon 19	c.3615C>G (p.Asn1205Lys)
12		Master. Kesavan T_12 - KAPPA BED		51x	1	0, 51	124x	1	0, 124	NAGLU	exon 6	c.1694G>T (p. Arg565Leu)
13		Master. Priyavarman_13 - KAPPA BED		96x	0.541667	44, 52	288x	0.440972	161, 127	MFN2	exon 6	c.545dupT (p.Met182IlefsTer6)
14		Master. Jagadish S_14 - KAPPA BED		29x	1	0, 29	208x	1	0, 208	SACS	Exon 10	c.4945A>G (p.Arg1649Gly)
15		Mrs. Munesswari_15 - MEDGENOME BED		39x	0.589744	16, 23	86x	0.325581	58, 28	DMGDH	Exon 1	c.97G>T (p.Glu337Ter)
16		Baby. Shravan Sudhan_16 - KAPPA BED		35x	0.514286	17, 18	150x	0.433333	85, 65	ITPR1	Exon 10	c.800C>T (p.Thr267Met)